

Executive Summary:

Automated Thyroid Nodule Segmentation Using Deep Learning Architectures

1. Introduction and Problem Statement:

Thyroid nodules are a common clinical finding, and their accurate localization and delineation in ultrasound images are essential for diagnosis, malignancy risk stratification, and treatment planning. Ultrasound imaging is widely used due to its safety, cost-effectiveness, and accessibility; however, manual segmentation of thyroid nodules is time-consuming and prone to inter- and intra-observer variability. These limitations motivate the development of automated, reliable, and reproducible segmentation methods to support clinical decision-making.

This project presents a deep learning–based framework for **automated thyroid nodule image segmentation**, focusing on a **comparative evaluation of three encoder–decoder convolutional neural network (CNN) architectures: U-Net, SegNet, and Residual U-Net**. The primary objective is to analyze how architectural design choices—such as skip connections, pooling strategies, and residual learning—affect segmentation accuracy, boundary delineation, and training stability in medical imaging applications.

2. Dataset and Preprocessing:

The experimental study was conducted using a curated thyroid ultrasound dataset consisting of **3,585 paired grayscale images and corresponding binary segmentation masks**, capturing variations in nodule size, shape, and texture. The dataset was split into **2,509 images for training** and **1,076 images for testing**, ensuring unbiased performance evaluation.

All images were standardized through preprocessing steps including grayscale normalization, intensity scaling, and resizing to **256 × 256 pixels**. These steps ensured consistent input representation across all models while preserving clinically relevant features necessary for accurate nodule segmentation.

3. Methodology and Model Architectures:

All three architectures were implemented using the **PyTorch framework**, and the segmentation task was formulated as a **pixel-wise binary classification problem**.

- **U-Net** was adopted as the baseline architecture due to its proven effectiveness in biomedical image segmentation. Its symmetric encoder–decoder design with skip connections enables multi-scale feature fusion, allowing precise recovery of spatial and boundary information.
- **SegNet** employs a memory-efficient decoding strategy by storing max-pooling indices during encoding and reusing them for upsampling via max-unpooling. While this approach preserves spatial alignment and reduces memory usage, the lack of direct feature concatenation limits its ability to reconstruct fine anatomical details.
- **Residual U-Net** extends the U-Net architecture by incorporating residual blocks within the encoder and decoder paths. These residual connections facilitate improved gradient flow, mitigate vanishing gradient issues, and enhance feature reuse, resulting in more stable training and robust segmentation performance.

All models were trained using the **Adam optimizer** with a learning rate of 1×10^{-4} , optimized using **Binary Cross Entropy (BCE) loss** over **25 epochs** in GPU-accelerated environments. Model performance was evaluated using the **Dice Coefficient**, a standard metric for assessing overlap accuracy in medical image segmentation.

4. Comparative Performance Analysis:

The quantitative comparison reveals clear performance differences among the three architectures, as summarized in **Table 1**.

Table 1: Comparative Performance of Segmentation Models

Model Architecture	Dice Coefficient	Training Stability	Boundary Delineation	Key Observations
U-Net	0.9765	Moderate	High	Excellent boundary capture; sensitive to hyperparameter tuning
SegNet	0.7783	High	Low	Memory efficient but poor fine-detail reconstruction
Residual U-Net	0.9732	Very High	Very High	Consistent performance with stable convergence

The **U-Net** achieved the highest peak Dice score, demonstrating strong capability in capturing nodule boundaries. However, its training process exhibited sensitivity to parameter tuning and stability variations.

SegNet showed significantly lower performance, particularly in boundary-sensitive regions, highlighting its limitations for ultrasound-based thyroid segmentation.

The **Residual U-Net** delivered near-peak accuracy with **superior training stability and consistency**, producing reliable segmentation outputs across training and testing phases.

5. Model Selection Rationale:

Although the standard U-Net achieved the highest single Dice score, the **Residual U-Net was selected as the optimal model** for practical deployment. In clinical settings, **robustness, reproducibility, and stable convergence** are often more critical than marginal improvements in peak accuracy. The Residual U-Net effectively balances high segmentation accuracy with reliable learning behavior, making it more suitable for real-world clinical integration.

6. Conclusion and Future Scope:

This project demonstrates that deep learning–based encoder–decoder architectures are highly effective for automated thyroid nodule segmentation from ultrasound images. The comparative analysis confirms that **architectural enhancements—particularly residual learning and multi-scale feature fusion—play a crucial role in improving segmentation reliability and robustness**. Among the evaluated models, the **Residual U-Net emerges as the most clinically viable solution**, offering consistent accuracy and stable performance.

Future work may extend this framework by integrating **attention mechanisms, multi-scale supervision, or transformer-based segmentation models**. Validation on larger, multi-institutional datasets and deployment within clinical imaging systems such as **computer-aided diagnosis (CAD) platforms or PACS** can further enhance the translational impact of this work.